

Instructions: There is totally 1 problem. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (10 points) Find the local maximum and minimum values and saddle points of the function,

$$f(x, y) = x^4 + y^4 - 4xy + 2.$$

$$\begin{cases} f_x = 4x^3 - 4y = 0 \\ f_y = 4y^3 - 4x = 0 \end{cases} \Rightarrow \begin{cases} x^3 - y = 0 \Rightarrow y = x^3 \\ y^3 - x = 0 \end{cases} \Rightarrow x^9 - x = 0$$

$$\Rightarrow x(x^8 - 1) = 0 \Rightarrow x = 0 \text{ or } x = 1 \text{ or } x = -1$$

$$\begin{matrix} \Downarrow & \Downarrow & \Downarrow \\ y = 0 & y = 1 & y = -1 \end{matrix}$$

Critical points: $(0, 0)$, $(1, 1)$, $(-1, -1)$

$$f_{xx} = 12x^2, \quad f_{yy} = 12y^2, \quad f_{xy} = -4.$$

$$D(x, y) = f_{xx}f_{yy} - f_{xy}^2 = 144x^2y^2 - 16$$

For $(0, 0)$

$$D = -16 < 0 \Rightarrow (0, 0) \text{ is a saddle point}$$

For $(1, 1)$

$$D = 144 - 16 = 128 > 0, \quad f_{xx} = 12 > 0$$

$$\Rightarrow (1, 1) \text{ is a local minimum, } f(1, 1) = 0$$

For $(-1, -1)$,

$$D = 144 - 16 = 128 > 0, \quad f_{xx} = 12 > 0$$

$$\Rightarrow (-1, -1) \text{ is a local minimum, } f(-1, -1) = 0.$$